Ryan M. Fraser

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Machine Learning Engineer and Data Scientist building production AI/ML systems.

Education

University of British Columbia — Vancouver, BC, Canada
Bachelor of Computer Science, GPA: 92% (A+) — *Sep. 2022 – Expected Dec. 2026*

- Dean of Science Scholar; Trek Excellence Scholarship; J. Fred Muir Memorial Scholarship

Work Experience

Machine Learning Software Engineer Intern — Vancouver, BC, Canada
RBC Borealis — *May 2026 – Present*

- Leading migration of business-metrics infrastructure to a new RBC Borealis cluster for a production multi-agent financial-advisor system.
- Built an aggregate-metrics pipeline with OpenSearch and Prometheus, surfaced via a Grafana dashboard.
- Migrated 5 data sources off third-party APIs using Adapter/Factory patterns, decoupling dashboards from source-specific logic.
- Took on core research duties after 2 researchers left, running monthly model evaluations and data-quality checks.

Software Developer Intern — Toronto, ON, Canada
RBC Amplify — *May 2025 – Aug. 2025*

- Designed and built a multi-agent orchestration system from scratch: a classifier agent routes user intent (data query, visualization, tagging) to specialized subagents executing tool calls over a unified de-normalized view aggregated from 5 internal data sources.
- Developed a text-to-SQL evaluation framework combining LLM-as-judge scoring with regex-based SQL validation (column presence, keyword correctness, injection detection), used to systematically compare prompt strategies and data pre-aggregation approaches.
- System deployed to 27+ analysts, reducing manual data gathering by ∼500 hours annually and contributing ∼$5M in business value.

Data Scientist Intern — Vancouver, BC, Canada
RBC Borealis — *Sep. 2024 – Apr. 2025*

- Built and optimized a random forest (LightGBM) model for residential home-price valuation, driving accuracy gains through feature engineering and 30+ experiments across feature-set variants.
- Verified the integrity and quality of both our team's and a partner team's datasets during exploratory analysis, surfacing data issues ahead of modeling.
- Benchmarked our model against the partner team's parallel valuation model and found their prediction errors were largely independent; ensembling the two improved prediction accuracy by 14%, and presented findings to stakeholders to support adopting the combined model.
- Prototyped a text-to-SQL pipeline for financial data, using LLM-as-judge evaluation to benchmark rule-based grammars against prompt-engineered LLMs and select the best query-generation strategy.

Technical Projects

Gaussian Mixtures For Golf Dispersion Patterns | *Python, Scikit-learn, NumPy, Jupyter, TypeScript*

- Modeled golfer shot distributions with Gaussian Mixture Models instead of the standard single-Gaussian dispersion ellipse, capturing multi-modal miss patterns a unimodal model cannot represent.
- Designed a model-selection pipeline (90/10 train-validation split) that filters the top-5 GMMs by energy distance then minimizes BIC to balance fit quality and complexity.
- Applied the fitted GMM as the transition model in a Markov Decision Process solved by value iteration, producing per-position expected-score heatmaps and optimal club/target recommendations per hole.

NVIDIA Nemotron LoRA Fine-tuning Kaggle Competition | *Python, Claude Code, PyTorch*

- Fine-tuned NVIDIA's Nemotron with a rank-32 Low Rank Adapter (LoRA) to improve performance on 7 types of logic puzzles.
- Augmented training data with Chain-of-Thought reasoning, teacher–student interactions, and prompt engineering, and applied a GRPO reinforcement-learning loop to improve reasoning and output formatting.

Technical Skills

Languages: Python, SQL, R, JavaScript, Java, C++
Machine Learning: PyTorch, Scikit-learn, LightGBM, Random Forests, Ensemble Methods, Feature Engineering, Pandas, NumPy, GMMs, Prompt Engineering, LLM Evaluation, Multi-Agent Systems
Data Engineering & Monitoring: PostgreSQL, OpenSearch, Airflow, Docker, Prometheus, Grafana, Git, Bash